

पी एम् श्री केन्द्रीय विद्यालय रानीखेत

Periodic Test-1 Class-12 Mathematics

TIME: 90 Min

M.M.40

General Instructions: All questions are compulsory.

Q.No	SECTION-A(1 Mark each)	Marks
1.	If X, Y and XY are matrices of order $2 \times 3, m \times n$ and 2×5 respectively, then find the number of elements in Y .	1
2.	Evaluate : $\tan^{-1} \left[2 \sin \left(2 \cos^{-1} \frac{\sqrt{3}}{2} \right) \right]$	1
3.	Let $S = \{1, 2, 3\}$ Determine whether the function $f: S \rightarrow S$ defined as below is one- one onto. $f = \{(1, 2), (2, 1), (3, 1)\}$	1
4.	Let A be a 3×3 matrix such that $ Adj A = 64$. Then find $ A $.	1
5.	Find the domain of $f(x) = \sin^{-1}(-x^2)$.	1
Q.No	SECTION-B (2 Mark each)	Marks
6.	Find k so that $f(x) = \begin{cases} \frac{x^2 - 2x - 3}{x + 1}, & x \neq -1 \\ k, & x = -1 \end{cases}$ Is continuous at $x = -1$	2
7.	Find the points on the curve $6y = x^3 + 2$ at which ordinate is changing 8 times as fast as abscissa.	2
8.	If $A = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$ and $A^2 = kA$, then find the value of k .	2
9.	Show that the function $f: R \rightarrow R$ defined by $f(x) = 4x^3 - 5, \forall x \in R$ is one-one and onto.	2
10.	Differentiate $\left(\frac{5^x}{x^5}\right)$ with respect to x .	2
11.	The volume of a cube is increasing at the rate of $6 \text{ cm}^3/\text{s}$. How fast is the surface area of cube increasing, when the length of an edge is 8 cm ?	2
Q.No	SECTION-C(3 Mark each)	Marks
12.	If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that $A^3 - 6A^2 + 7A + 2I = 0$.	3
13.	Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by $R : (a, b) R(c, d) : ad(b + c) = bc(a + d)$. Show that R is an equivalence relation	3
14.	A furniture workshop produces three types of furniture – chairs, tables and beds each day. On a particular day the total number of furniture pieces produced is 45. It was also found that production of beds exceeds that of chairs by 8, while the total production of beds and chairs together is twice the production of tables. Determine the units produced of each type of furniture, using matrix method.	3

Q.No	SECTION-D (4 Mark each)	Marks
15.	If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.	5
16.	If $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 0 & 2 \\ -3 & -3 & 0 \end{bmatrix}$. Find A^{-1} and hence solve the equations. Using this solve the system of linear equations. $x + 2y - 3z = 1$, $2x - 3z = 2$, $x + 2y = 3$	5
SECTION-E (CASE BASED PROBLEM)		
17.	 A small town is analysing the pattern of a new street light installation. The lights are set up in such a way that the intensity of light at any point x metres from the start of the street can be modelled by $f(x) = e^x \sin x$, where x is in metres. Based on the above, answer the following: (i) Find the intervals on which the $f(x)$ is increasing or decreasing, $x \in [0, \pi]$. (ii) Verify whether each critical point when $x \in [0, \pi]$ is a point of local maximum or local minimum or a point of inflexion.	2 2